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B.Tech. Degree VI Semester Regular Examination April 2022

ME 19-205-0602 MACHINE DESIGN I (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Understand the customers' need, formulate the problem and draw the design specifications.

CO2: Understand component behaviour subjected to loads and identify the failure criteria.

CO3: Apply the concepts of stress analysis, theories of failure and material science to analyse, design and/or select commonly used machine components.

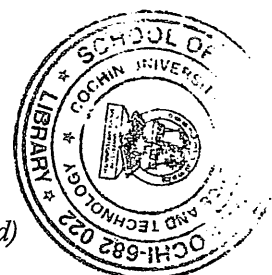
CO4: Analyse the stresses and strains induced in a machine element.

CO5: Design machine components using theories of failure.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze,

L5 – Evaluate, L6 – Create

PO – Programme Outcome

*(Use of Design Data Handbook is permitted)**(Data not given may be suitably assumed. All such assumptions must be clearly stated)*

PART A

(Answer ALL questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
I.						
(a)	Discuss the statement "In static loading stress concentration in ductile materials is not as serious as in brittle material". Then why the effect of stress concentration effect has to be considered for both materials in fatigue loading?		3	L3	2	1,2
(b)	What are the factors to be considered for selection of material used in sports equipments and for aerospace applications? Briefly explain.		3	L2	1	1,2
(c)	List and explain the types of Keys.		3	L1	1	1
(d)	What are the requirements of a good shaft coupling?		3	L2	2	1,2
(e)	Explain in context of riveting:		3	L1	1	1,2
	(i) Pitch					
	(ii) Back pitch					
	(iii) Diagonal pitch					
	(iv) Margin.					
(f)	Explain Nipping in leaf springs.		3	L2	2	1,2
(g)	List the advantages and disadvantages of welded joints over riveted joints.		3	L2	2	1
(h)	What are the stresses that are induced in the shafts? Briefly explain.		3	L2	2	1,2

PART B

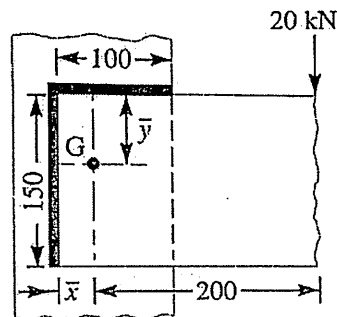
		(4 × 12 = 48)				
II.	A pulley is keyed to a shaft midway between two bearings. The shaft is made of cold drawn steel for which the ultimate strength is 550 MPa and the yield strength is 400 MPa. The bending moment at the pulley varies from -150 N-m to +400 N-m as the torque on the shaft varies from -50 N-m to +150 N-m. Obtain the diameter of the shaft for an indefinite life. The stress concentration factors for the keyway at the pulley in bending and in torsion are 1.6 and 1.3 respectively. Take the following values: Factor of safety = 1.5, Size effect factor = 0.85, Surface effect factor = 0.88		12	L5	3	2,3

OR

(P.T.O)

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|-----------|--|----|----|---|-------|
| III. (a) | A cylindrical shaft made of C 40 steel is subjected to static loads consisting of bending moment 10 kNm and a torsional moment 30 kNm. Determine the diameter of the shaft using any two different theories of failure, and assuming a factor of safety of 2. | 6 | L3 | 5 | 1,2,3 |
| (b) | A cantilever beam 300 mm length of circular cross section of diameter 60 mm is subjected to a load of 2 kN at its free end. The beam is also subjected to the torsional moment of 1.5 kNm. Determine the maximum principal and shear stress. | 6 | L3 | 4 | 2,3 |
| IV. | Two ends of a pump are joined by means of a cotter joint at the ends. Design the joint for an axial load of 100 kN. The allowable stresses for the material used are 50 MPa in tension, 40 MPa in shear and 100 MPa in crushing. | 12 | L4 | 3 | 2,3 |
| OR | | | | | |
| V. | Design a flanged coupling to transmit 18 kN at 1440 rpm. The allowable shear stress for the cast iron flange is 4 MPa. The shaft, key and bolts are made of annealed steel having allowable shear stress of 93 MPa. Allowable crushing stress for the key is 186 MPa. | 12 | L4 | 3 | 1,2,3 |
| VI. | Design a triple riveted double strap butt joint with chain riveting for a boiler of 1.5 m diameter and carrying a pressure of 1.2 N/mm ² . The allowable stresses are:
$\sigma_t = 105$ MPa, $\tau = 77$ MPa and $\sigma_c = 162.5$ MPa | 12 | L3 | 3 | 2,3 |
| OR | | | | | |
| VII. | A multi leaf spring is fitted to the chassis of an automobile over a span of 1.25 m to absorb the shock due to maximum load of 24 kN. All the leaves are to be stressed to 540 MPa, when fully loaded. The spring has 2 full length leaves and 6 graduated leaves with a central band of 100 mm width. The maximum deflection is 90 mm. Determine the width and thickness of the leaf. Take $E = 206$ GPa. | 12 | L4 | 5 | 2,3 |
| VIII. | Determine the size of the weld for plate welded as shown in figure. Take permissible shear stress for the weld material as 80 Mpa. | 12 | L4 | 3 | 1,2,3 |



All dimensions in mm.

OR

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|---------|---|---|----|---|-----|
| IX. (a) | A shaft made of mild steel is required to transmit 100 kW at 300 rpm. The supported length of the shaft is 3 meters. It carries two pulleys each weighing 1500 N supported at a distance of 1 metre from the ends respectively. Assuming the safe value of stress, determine the diameter of the shaft. | 8 | L3 | 5 | 2,3 |
| (b) | Discuss the advantages and disadvantages of a hollow shaft in comparison to a solid shaft. | 4 | L2 | 2 | 1 |

Bloom's Taxonomy Levels:

L1 – 7.14% , L2 – 20.5% , L3 – 27.6% , L4 – 35.7% , L5 – 8.90% , L6 – 0%
